

The Effect of Typhoons on Vegetable Prices and Quantities in Taiwan¹

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Abstract

Taiwan experiences several typhoons each summer, and these storms often cause sharp but short-lived disruptions in the vegetable market. This paper estimates the effect of typhoons on vegetable prices and quantities traded using daily market data from the Taipei wholesale market in 2023. Applying an event-study OLS regression with typhoon dummy variables for five days centred on each storm event, we find that prices rise by about 15% and quantities fall by about 9% on the typhoon day. The price increase begins one day before the storm—consistent with precautionary purchasing by consumers responding to official warnings—and both prices and quantities return to normal within two days after the storm passes.

Keywords: typhoon, vegetable prices, Taiwan, event study, OLS

¹This paper is prepared solely as a format example for student guidance. All data, estimation results, and references are fictitious and do not represent real empirical findings. For the actual empirical analysis, see <https://jiandazhu.github.io/doc/Typhoon.pdf>.

1 Introduction

Taiwan lies on the main track of Pacific typhoons and is struck by several storms every summer, typically between July and October. Typhoons bring heavy rainfall and strong winds that destroy crops, wash out farm roads, and disrupt the supply chain from fields to wholesale markets, often causing sharp spikes in fresh-vegetable prices (Chen, 2022; Johnson and S. K. Chen, 2018). Understanding the size and timing of these price changes matters for policymakers: targeted responses—such as releasing emergency reserves or issuing supply reassurances—can help protect households, especially lower-income families, from sudden food-price increases. This study uses market data to investigate how typhoons affect vegetable prices and quantities traded in Taiwan.

We address this question using daily market data for four vegetable categories at the Taipei First Wholesale Vegetable Market in 2023. We estimate an event-study OLS regression that includes dummy variables for the two days before, the day of, and the two days after each typhoon event, controlling for vegetable-category and month fixed effects, following the approach of Yu, McCarl, and Zhu (2022).

The results show that prices rise by roughly 15% and quantities fall by roughly 9% on the typhoon day. Prices begin to increase one day before the storm, consistent with precautionary purchasing after official warnings are issued, and both series return to normal within two days after the storm passes.

2 Data and Methodology

2.1 Data

We use daily transaction data from the Taipei First Wholesale Vegetable Market covering 1 January to 31 December 2023. Following the classification of Taiwan’s Council of Agriculture, the data cover four vegetable categories: *leafy vegetables* (e.g., bok choy, cabbage, spinach), *flower and fruit vegetables* (e.g., eggplant, cucumber, bitter melon), *root vegetables* (e.g., radish, carrot, onion), and *mushrooms* (e.g., shiitake, oyster mushroom). Each category provides one observation per day, yielding 1,460 observations in total (4 categories $\times$ 365 days).

The sample period contains four typhoon land-warning events, which provide the identifying variation for our analysis. These events occurred in late July, twice in August, and in mid-September 2023, producing eight land-warning calendar days and 32 typhoon-day observations across the four vegetable categories.

Monthly average prices display a clear seasonal spike during the typhoon season. As Figure 1 shows, prices rise sharply between July and September and return to near-baseline levels by October, consistent with the timing of the four storm events.

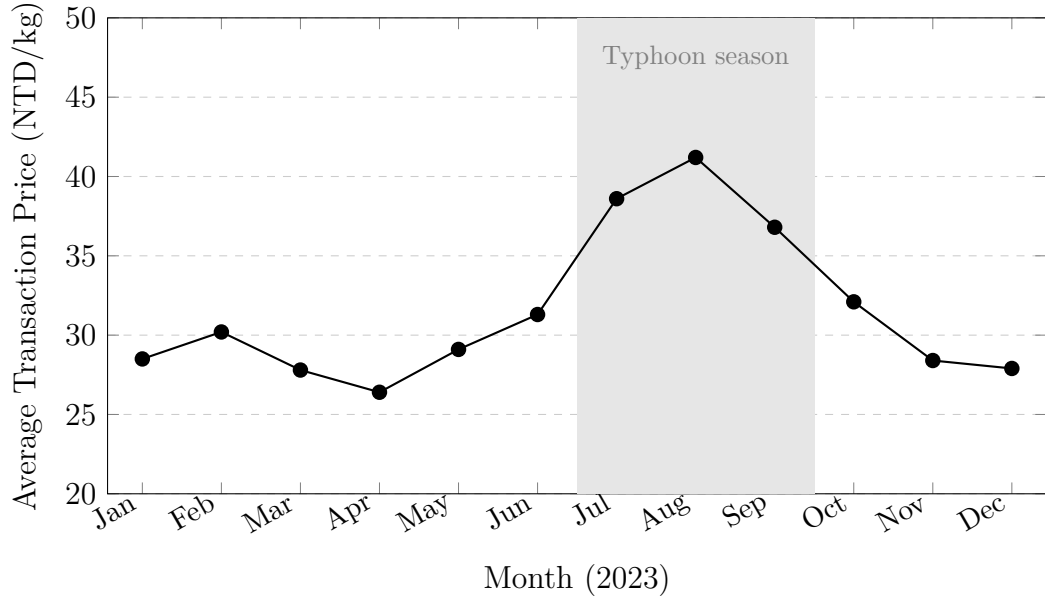


Figure 1: Monthly average vegetable prices in 2023. The shaded area indicates the typhoon season (July–September).

Raw comparisons confirm that typhoon days are associated with substantially different market conditions. Table 1 reports summary statistics for the full sample (Panel A) and separately by typhoon status (Panel B); on typhoon days, average prices are notably higher (NTD 46.2/kg vs. NTD 29.1/kg) and quantities notably lower (3,180 kg vs. 4,280 kg).

Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Panel A: Full sample</i>					
Price (NTD/kg)	1,460	29.8	8.4	8.5	78.6
Quantity (kg)	1,460	4,250	1,820	320	18,600
Typhoon day (= 1)	1,460	0.022	0.146	0	1
<i>Panel B: By typhoon status</i>					
<i>Non-typhoon days</i>					
Price (NTD/kg)	1,428	29.1	8.2	8.5	75.3
Quantity (kg)	1,428	4,280	1,750	350	18,600
<i>Typhoon days</i>					
Price (NTD/kg)	32	46.2	12.8	28.4	78.6
Quantity (kg)	32	3,180	980	320	5,240

Note: Each observation is one vegetable category on one day. “Typhoon day” equals 1 on days when a typhoon land warning is in effect.

2.2 Empirical Model

To identify the effect of typhoons on the vegetable market, we estimate:

$$\log(Y_{it}) = \alpha + \sum_{k=-2}^{+2} \beta_k D_{k,t} + \gamma_i + \delta_m + \varepsilon_{it}, \quad (1)$$

where Y_{it} is either the price or the quantity traded for vegetable category i on day t ; $D_{k,t}$ equals 1 if day t falls exactly k days relative to a typhoon event ($k \in \{-2, -1, 0, +1, +2\}$) and 0 otherwise; γ_i is a vegetable category fixed effect that absorbs permanent price differences across categories; δ_m is a month fixed effect that absorbs seasonal patterns; and ε_{it} is the error term.

The coefficients β_k capture the percentage change in Y on day k relative to a typhoon compared with ordinary non-typhoon days. Because typhoon timing is determined by meteorological conditions independent of vegetable market conditions, we treat $\hat{\beta}_k$ as causal estimates (Yu, McCarl, and Zhu, 2022). Standard errors are clustered at the vegetable category level.

3 Results

Table 2 reports estimates of equation (1). Columns (1)–(2) use log price as the dependent variable; columns (3)–(4) use log quantity. Odd-numbered columns include no fixed effects; even-numbered columns add vegetable-category and month fixed effects.

The results are consistent across specifications. Two days before the typhoon (D_{-2}), prices and quantities show no significant change, confirming that the market is undisturbed at this horizon. One day before (D_{-1}), prices rise by approximately 8%—a pre-typhoon increase consistent with consumers making precautionary purchases after official warnings are issued (Lee, Wang, and Lin, 2021)—while quantities fall slightly. On the typhoon day (D_0), prices peak at around 15% above normal and quantities drop by roughly 9%, reflecting supply disruptions as typhoon conditions interrupt harvesting and transportation (Martinez and Tanaka, 2019). By two days after the storm (D_{+2}), both prices and quantities have returned to near-normal levels, suggesting swift market recovery.

Comparing columns (1) and (2)—or (3) and (4)—shows that adding fixed effects leaves the typhoon coefficients almost unchanged, while raising R^2 substantially. This indicates that the event-study estimates are not confounded by seasonal patterns or persistent category-level price differences.

Table 2: Effect of Typhoons on Vegetable Prices and Quantities

	log(Price)		log(Quantity)	
	(1)	(2)	(3)	(4)
D_{-2} : 2 days before	0.031 (0.024)	0.028 (0.022)	0.009 (0.018)	-0.012 (0.016)
D_{-1} : 1 day before	0.082** (0.031)	0.079** (0.029)	-0.021* (0.012)	-0.024** (0.011)
D_0 : Typhoon day	0.152*** (0.038)	0.148*** (0.035)	-0.086*** (0.021)	-0.091*** (0.019)
D_{+1} : 1 day after	0.094** (0.034)	0.091** (0.032)	-0.063*** (0.018)	-0.067*** (0.016)
D_{+2} : 2 days after	0.018 (0.022)	0.016 (0.021)	0.008 (0.014)	0.005 (0.013)
Category FE	No	Yes	No	Yes
Month FE	No	Yes	No	Yes
N	1,460	1,460	1,460	1,460
R^2	0.158	0.684	0.112	0.612

Note: Robust standard errors in parentheses, clustered at the vegetable category level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4 Conclusion

This paper estimates the effect of typhoons on Taiwan’s wholesale vegetable market using daily market data for 2023. We find that on the typhoon day, prices rise by about 15% and quantities fall by about 9%, with a pre-typhoon price increase one day before landfall pointing to precautionary purchasing by consumers responding to official warnings (Lee, Wang, and Lin, 2021). Both series recover to near-normal levels within two days.

These findings suggest two policy responses. First, the government could issue timely supply reassurances alongside typhoon warnings to reduce unnecessary panic buying. Second, maintaining emergency vegetable reserves for rapid release during and immediately after typhoons would help stabilise prices and protect household food budgets. Future research using actual market data, multiple markets, and typhoon-intensity measures would allow for a more precise assessment of these effects (Yu, McCarl, and Zhu, 2022).

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